MILITARY SPECIFICATION

STARTER, PNEUMATIC, AIRCRAFT ENGINE
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-S-38399A(USAF), dated 14 February 1975, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, following "STANDARDS, MILITARY": Add:

"PUBLICATIONS

AIR FORCE SYSTEMS COMMAND DESIGN HANDBOOK

AFSC DH 1-2 General Design Factors".

PAGE 6

3.7.1.4, Insert following first sentence:

"The release of sparks from the starter exhaust is permissible. Precautions in addition to the preceding that are required of the starter to prevent a hazardous condition to the aircraft shall be specified in the detail specification."

PAGE 8

3.7.5.3, Add the following last sentence:

"Fasteners and fittings shall be in accordance with AFSC Design Handbook 1-2, Chapter 4, with the exception that Helicoil inserts will not be used in magnesium where installation and removal of the fastener are likely to occur many times during the life of the equipment."

PAGE 12

3.10.3, Insert at the beginning of first sentence: "Unless otherwise specified in the detail specification..."

FSC 2995

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PAGE 14

3.11.9, Delete and substitute:

"3.11.9 Attitude

a. The starter shall meet the overrunning requirements of this specification in the following attitudes:

Continuous overrunning: 0° to 45° dive or climb combined with 10° roll in either direction.

Intermittent overrunning: 45° to 105° nose up position for 3 minutes, 45° to 90° nose down position for 3 minutes.

b. The starter shall meet the starting requirements of this specification in attitudes from 0° to 45° dive combined with 10° of roll in either direction and 20° climb combined with 10° roll in either direction."

3.12.1.1.5, Delete second, third, and fourth sentences.

PAGE 16

3.12.4, Delete the last sentence and substitute:

"Unless otherwise specified in the detail specification the starter shall be designed for 1,000 hours of overrunning without changing or adding to the lubricant."

PAGE 19

4.3.6.3, Delete and substitute:

"4.3.6.3 Test flywheels. For the preproduction test, either a single flywheel in conjunction with a test stand loading system or two flywheels may be used for loading the starter. If the single flywheel and test stand loading system are used, both normal and rapid acceleration schedules shall be simulated. If two flywheels are used, they shall have different moments of inertia as described below and shall be referred to as flywheel A and flywheel B."

PAGE 20

4.4.2.1 b. Delete.

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PAGE 21

4.4.2.3 b. Delete "250 cycles minimum" and substitute: "1000 cycles minimum".

4.4.2.3 d. Delete "250 cycles minimum" and substitute: "1000 cycles minimum".

4.4.3.1. Delete last sentence and substitute:

"Unless otherwise specified in the detail specification, the oil change interval shall not be less than 1000 hours of overrunning."

PAGE 22

4.4.4 b. Add to the end of the sentence:

"Unless the design is changed, then 4.4.5 governs."

4.4.5, second line: Delete "stics".

PAGE 23

4.5.3, Delete the last sentence and substitute:

"Once a lot has been rejected, full particulars concerning the cause of rejection and the action(s) necessary to check for and correct as necessary the defects in the lot shall be furnished in writing to the procuring activity."

4.6.5 a. Delete.

PAGE 24

4.6.5 c. Delete "1 minute" and substitute "10 seconds".

4.6.5 d. Delete.

PAGE 25

4.6.10, first sentence: Delete and substitute:

"Unless otherwise specified in the detail specification, the starter shall be subjected to vibration test in accordance with MIL-STD-810, Method 514.2 equipment Category b.1 to the acceleration levels of figure 514.2-2, curve L, mounted without vibration isolators and mounted directly on the aircraft equipment that duplicates the vibration characteristics of the aircraft."

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PAGE 27

4.6.15: Add at the end of paragraph,

"During the running engagement test, one series of engagements shall be made at each of the following attitudes:

45° dive 10° left roll
45° dive 10° right roll
20° climb 10° left roll
20° climb 10° right roll

PAGE 28

4.6.18, Delete and substitute:

"4.6.18 Clutch and shaft failure test. The starter shall be installed on a test stand with a weakened shear shaft installed. A start cycle shall be run that will cause the shaft to shear and the unit to shut down in its normal manner. A second start cycle shall then be performed with the shaft sheared. Any damage to the starter or the mounting pad shall be cause for rejection. If the starter incorporates an emergency disengagement feature, the following test should also be run. The starter shall be disassembled sufficiently to render the clutch inoperative. After reassembly, the output shaft shall be accelerated until the emergency disengagement feature is actuated. The starter shall then be subjected to a normal acceleration cycle with the disengagement feature actuated. Damage to the starter shall be cause for rejection."

PAGE 33

6.3.1, Delete and substitute:

"6.3.1 Procuring activity. The term procuring activity as used in this specification is the agency purchasing the starter and may be other than the Government.

PAGE 35

Add:

"6.3.17 Lot. A lot is a collection of units of products bearing identification and treated as a unique entity from which a sample is to be drawn and inspected to determine conformance with the acceptability criteria."

6.3.18 Failure. A failure is defined as any malfunction which requires adjustment, repair, replacement or maintenance in order to restore the starter to operational status."

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Reviewer activity: Delete "71" and substitute "99".

Custodian:
Air Force - 11

Reviewer activity:
Air Force - 99

Preparing activity:
Air Force - 11

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